

A Transfer Learning Based Approach to Facial Emotion Recognition

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Introduction

- In Facial Emotion Recognition (FER), a machine learning model is trained to classify presented faces into emotional categories.
- This type of model has several applications such as in security, marketing and healthcare.
- Current approaches to FER primarily use Convolutional Neural Networks.
- The goal of this study was the make improvements to the current state of the art using transfer learning.

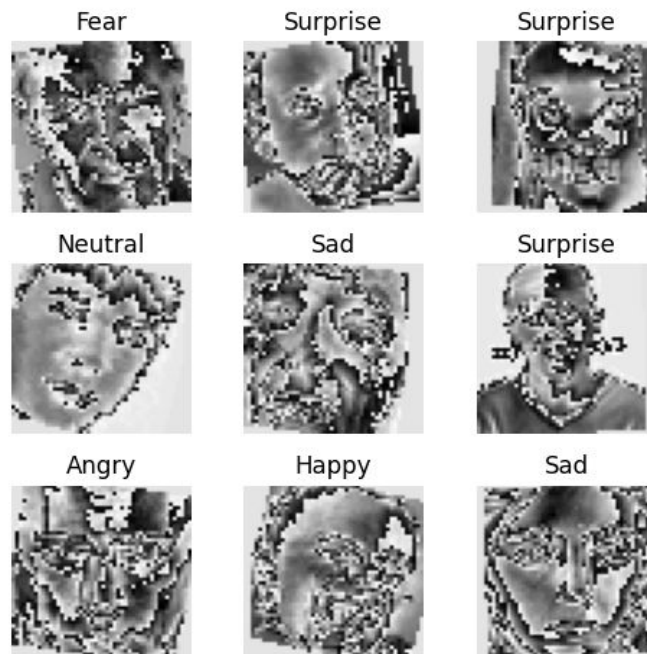
Methods

- The pre-trained model ResNet-18 was run on the FER 2013 and CK+ datasets (consisting of 48x48 grayscale faces and emotional labels).
- The faces were normalised with respect to illumination and augmented through random cropping and horizontal flipping prior.
- Cross entropy loss was minimised in order to train the model.
- Learning-rate adaptation and hyperparameters were adjusted to improve model performance.
- An ensemble learning approach was also attempted.

FER2013 Dataset

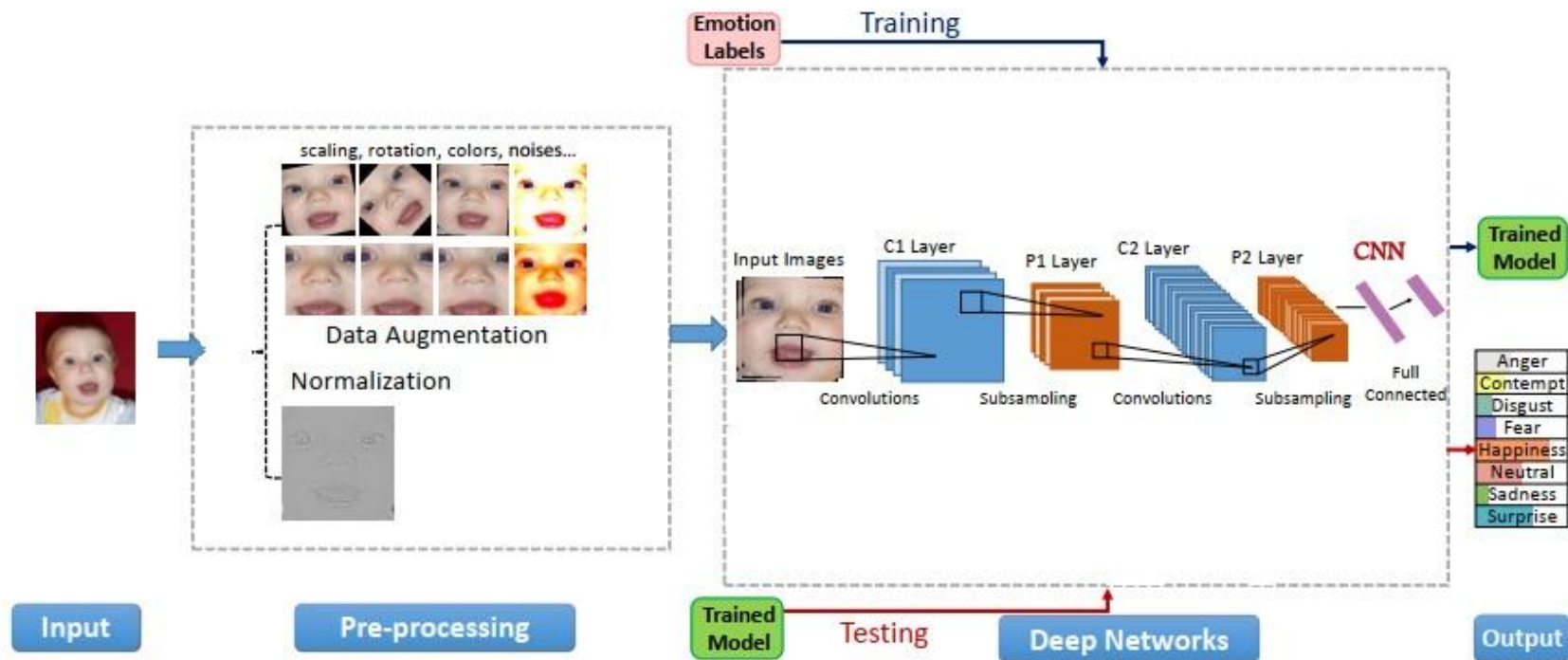


Input images



Preprocessed images

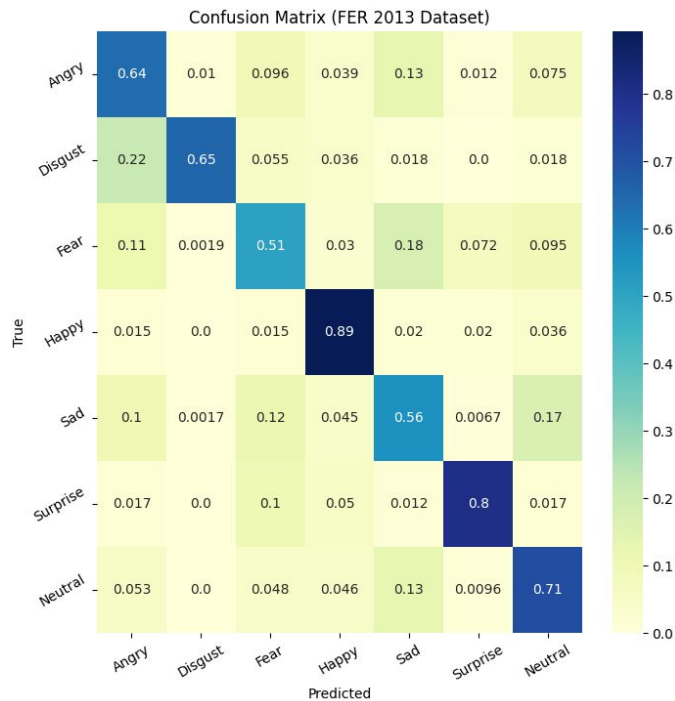
Model



Results: “Accuracy”

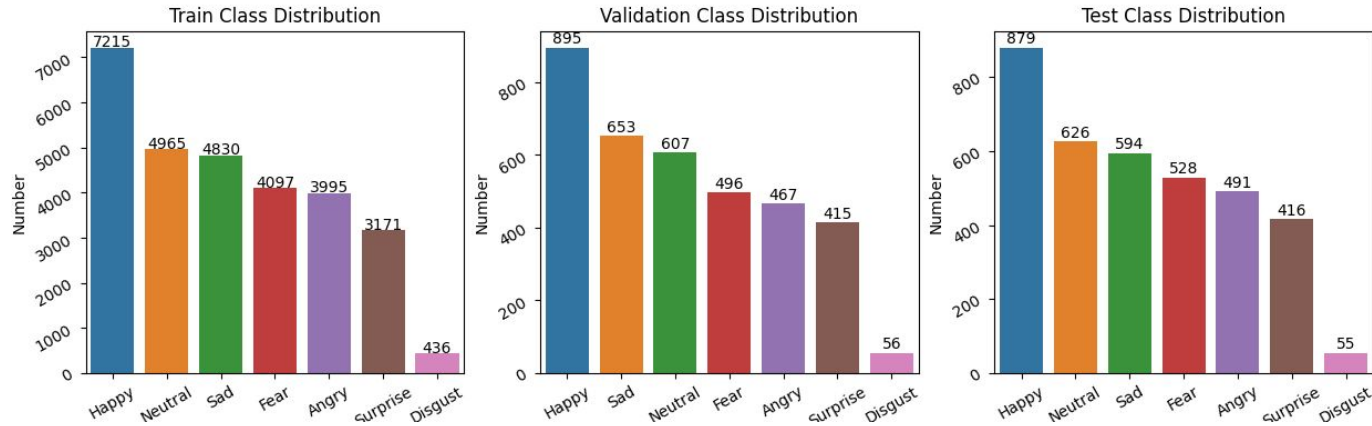
#	Method/ Model	Train Acc(%)	Validation Acc(%)	Test Acc(%) FER2013	Test Acc(%) CK+	Learning rate	Optimizer	Batch size	Epochs	No. of Trainable Parameters
1	ResNet18 Transfer Learning	31.04	31.64	31.98	-	Adaptive (StepLR)	Adam	256	10	3.5K
2	S-ResNet18	87.13	65.86	67.26	67.58	CosineAnneal LR	Adam	128	50	11M
3	Feature_Level-Ensemble-Learning	82.3	64.92	66.17	59.42	CosineAnneal LR	SGD	64	20	132K
4	ResNet18 Fine Tuning	86.86	69.29	70.10	68.70	Adaptive (StepLR)	Adam	128	20	11 M

Results: “Confusion Matrix”



Discussion

- The model performance is encouraging but not upto the current standard.
- This could be explained by limitations in the image resolution and the number of trainable parameters in our models.
- The lower accuracy for certain emotions could be explained by fewer faces with that label in the datasets.



Thank you!